

1.3.1 – 1.3.4 Communication Models & OSI 7-Layer Model

کمیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔

CS XI

Chapter 1: Computer Systems

Lecture 11 of 12

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1.3.1 – 1.3.4 Communication Models & OSI 7-Layer Model

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LEARN IT

A **communication model** explains how data travels from one device to another through a network. Because sending data is complicated, the job is split into **layers**; each layer does one part of the job and passes the data to the next layer.

The **OSI (Open Systems Interconnection) model** was developed by **ISO** (International Organization for Standardization) in **1984**. It divides communication into **7 layers**. When you send data, it goes **down** the layers on your device (7 → 1), travels as bits over the cable or air, and goes **up** the layers on the receiving device (1 → 7).



The 7 layers of the OSI model with the mnemonic "Please Do Not Throw Sausage Pizza Away".

REMEMBER IT

Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top to bottom you can use "All People Seem To Need Data Processing".)

OSI layers — top layers (7 to 4)

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)

OSI layers — bottom layers (3 to 1)

Layer	Function	Protocols / hardware
3. Network	Logical addressing (IP addresses) and routing packets across networks.	IP, ICMP (Ping), Routers
2. Data Link	Physical addressing (MAC addresses) and error detection in frames.	Ethernet, MAC, Switches
1. Physical	Transmits raw bits (0, 1) over cables or wireless media.	Cables, Hubs, Wi-Fi radio

TCP vs UDP at the Transport layer

	TCP	UDP
Main feature	Reliable — checks that all data arrives correctly and in order.	Fast — sends data without waiting for checks.
Use when	Accuracy matters.	Speed matters more than perfection.
Examples	Web browsing, emails	Video streaming, online gaming

WORKED EXAMPLE

Tracing an email through the OSI layers (sender side).

Layer 7 (Application): you press Send; your email program uses **SMTP**.

Layer 6 (Presentation): the text is encoded (ASCII) and may be encrypted with TLS.

Layer 5 (Session): a session is opened with the mail server.

Layer 4 (Transport): **TCP** breaks the data into segments and numbers them, so nothing is lost (email needs accuracy).

Layer 3 (Network): each packet gets the **IP address** of the destination; **routers** choose the path.

Layer 2 (Data Link): packets are put in **frames** with **MAC addresses**; a **switch** forwards them on the local network.

Layer 1 (Physical): the frames are sent as raw **bits** over the cable or Wi-Fi radio.

At the receiver, the same steps happen in reverse order (1 → 7).

COMMON MISTAKE

Do not mix up the two addresses: **IP address** = logical address = **Network layer (3)**; **MAC address** = physical address = **Data Link layer (2)**. Also, routers work at layer 3, switches at layer 2 and hubs at layer 1.

اردو میں سمجھیں

کمیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔ او ایس آئی (OSI) ماڈل 1984 میں آئی ایس او (ISO) نے بنایا اور اس کی سات تہیں (Layers) ہیں: فزیکل، ڈیٹا لنک، نیٹ ورک، ٹرانسپورٹ، سیشن، پریزنٹیشن اور ایپلی کیشن۔ ہر تہ کا الگ کام ہے، جیسے نیٹ ورک تہ IP ایڈریس اور روٹنگ سنبھالتی ہے۔ TCP درستگی کے لیے اور UDP رفتار کے لیے استعمال ہوتا ہے۔

EXAM TIP

A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two protocols/devices for each layer. MCQs often ask: "Which layer handles routing?" (Network), "Which layer does encryption?" (Presentation), "Who developed OSI and when?" (ISO, 1984), "Which protocol is used for video streaming?" (UDP).

CHECK YOURSELF

1. Which OSI layer is responsible for encryption and compression?
2. At which layer do switches and MAC addresses work?
3. Should online gaming use TCP or UDP? Why?

Answers: 1. Presentation layer (6). 2. Data Link layer (2). 3. UDP, because speed matters more than perfect accuracy.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. OSI has:

- a. 7 layers
- b. 4 layers
- c. 2 layers
- d. 12 layers

MCQ 2. Layer 7 is:

- a. Application
- b. Physical
- c. Network
- d. Data link

MCQ 3. IP addressing lives at:

- a. Network (layer 3)
- b. Application
- c. Physical
- d. Session

MCQ 4. Cables and bits are:

- a. Physical
- b. Presentation
- c. Session
- d. Application

MCQ 5. End-to-end ports (TCP/UDP) are:

- a. Transport
- b. Network
- c. Physical
- d. Data link

MCQ 6. The model that explains “how data travels” is a:

- a. Communication model
- b. Sort algorithm
- c. K-map
- d. MVP

Extra short questions

1. Write the 7 OSI layers from top to bottom.
2. One job of Transport and of Network.
3. Why layer models?
4. Mnemonic you will actually write in the exam.

Extra long question

1. Name all 7 OSI layers with one job and one protocol/hardware example each. ★
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Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Application, Presentation, Session, Transport, Network, Data Link, Physical.
2. Transport: reliability/ports (TCP/UDP). Network: routing and logical addressing (IP).
3. Each layer has a small job; vendors can change one layer without rewriting the rest.
4. Please Do Not Throw Sausage Pizza Away (bottom-up) or All People Seem To Need Data Processing (top-down).

Long

1. 7 App HTTP/DNS; 6 Pres encryption/JPEG; 5 Sess dialogs; 4 Trans TCP/UDP; 3 Net IP/routers; 2 DL frames/MAC/switches; 1 Phy cables/bits/hubs. Draw the stack.

Next lecture: 1.3.5 – 1.3.6 TCP/IP Model vs OSI Model